From pairs trading to cost-aware relative value

Literature review, mathematical design and audited backtests

Prepared for Fakhruddin Hakim Hussain | 25 October 2026

The research now has a defensible economic mechanism and reproducible positive results under declared assumptions. The improvement comes from deciding when carry pays for capital and execution, rather than adding a more complex classifier.

What the evidence supports

Keep the cost-aware carry engine as the strongest extension of the original project. Retire the old notebook's unverified performance claims. Treat the result as a research prototype: excess returns are modest and concentrated, and execution/counterparty risks remain outside the historical price ledger.

Funding carry | Jan 2023-Aug 2026

Model	Net total	Net CAGR	Daily excess SR
Always on	13.56%	3.53%	-3.94
EWMA gate	18.97%	4.85%	2.22
Cash proxy	17.52%	4.50%	-

Dated futures | Jan 2023-Jun 2026

Model	Net total	Net CAGR	Daily excess SR
Always on	16.81%	4.55%	0.01
Cost + cash gate	19.35%	5.20%	0.93
Cash proxy	16.79%	4.55%	-

All costs included. Cash proxy: lagged SOFR minus 25bp, floor zero. Total returns include cash interest. Different end dates mean these two experiments are not directly ranked. Read daily Sharpe with the dependence diagnostics in section 6.

1 | What changed in the equity idea

The original result was not an investable backtest

The original notebook fitted PCA, clustering and pair tests across the same 2015-2019 history it traded. It then kept pairs that outperformed on that history. Signals and fills used the same close, while the ledger omitted daily unrealised risk, trading costs and short borrow. The fitted cointegrating spread was replaced by a ratio at the trading stage.

Other audit findings: a list was modified during iteration; the notebook retained 39 winners while the README said 38; the reviewed code did not calculate the CV Sharpe of 1.09. The frozen equity universe, unclear corporate-action adjustments and absent borrow availability prevent a reliable overnight NSE-short claim.

A causal statistical-convergence pilot

A fresh pilot used 16 economically related ETF pairs. Estimates used the prior 504 trading sessions and refitted quarterly. Trading used next-open prices, whole shares, cash distributions, borrow charges and a cash benchmark. It compared rolling spread signals, an OU cost gate and a partial-cointegration cost gate.

The state-space model separated a temporary AR(1) deviation from a permanent random walk. Filtered states used only available observations; smoothed states were excluded from signals. The hedge fit was conditional, rather than the authors' joint estimator.

Component	Research purpose
Temporary state M(t) = rho M(t-1) + noise	Estimate the displacement that may reverse.
Permanent state R(t) = R(t-1) + noise	Avoid assuming every shift must reverse.
Expected convergence versus costs	Require enough predicted return to pay for implementation.

Validation rejected this implementation

All three ETF variants had negative net excess Sharpe in 2019-2021 at their base costs. The later ETF strategy test was left unopened. This is a useful negative result: statistical fit alone did not establish a tradable edge.

The stronger question is therefore economic: what forces or pays for convergence? For spot-futures pairs, both legs reference the same asset. Funding and fixed maturity give measurable mechanisms that do not rely on two separate companies remaining economically similar.

2 | Literature translated into decisions

The review uses original papers, author implementations and official contract documentation. Access limitations are recorded rather than presenting every item as a full-paper review. The separate Markdown review gives the fuller synthesis.

Source	Decision carried into this project
Gatev, Goetzmann & Rouwenhorst (2006)	Separate historical pair formation from subsequent trading. Distance rules are benchmarks, not proof of a present-day edge. Access: Abstract/working-paper summary.
Avellaneda & Lee (2010; 2009 manuscript)	Model factor-adjusted residuals rather than arbitrary price ratios; inspect changing regimes and implementation costs. Access: Full author manuscript.
Clegg & Krauss (2018)	Allow temporary mean reversion alongside permanent relationship shifts. Our conditional hedge fit is simpler than their joint procedure. Access: Abstract + author implementation.
Leung & Li (2015)	Entry and liquidation depend on costs and stopping rules. Our cost gate is a heuristic, not their optimal-stopping solution. Access: Full author preprint.
Rad, Low & Faff (2016)	Compare simple and complex dependence models after time-varying costs. More elaborate models need empirical justification. Access: Full accepted manuscript.
Bailey & Lopez de Prado (2014)	Record trials and selection. A best observed Sharpe is inflated by research choices; no automatic correction erases all discretion. Access: Full author manuscript.
Engelberg, Gao & Jagannathan	Temporary liquidity pressure differs from permanent information. Price filters do not reveal the cause of a shock without news data. Access: Abstract only.
Schmeling, Schrimpf & Todorov (BIS, 2023)	Carry compensates constrained arbitrage capital. Dated-futures basis has a different convergence mechanism from perpetual funding. Access: Full BIS paper.
He, Manela, Ross & von Wachter	Perpetual pricing depends on funding rules, costs and arbitrage capacity. Positive funding is not a fixed-maturity payoff guarantee. Access: Full-text preprint HTML.
Ackerer, Hugonnier & Jermann	Contract denomination and funding specifications matter; linear and inverse contracts require different cash-flow ledgers. Access: Full-text preprint HTML.
Lo (2002)	Serial dependence changes Sharpe annualisation. Report a long-run-variance diagnostic and block uncertainty alongside the usual daily statistic. Access: Publication abstract.

3 | Dated futures: match the coin hedge

Research question

Can a visible futures premium compensate for execution costs and the return available on cash, while the purchased coin itself collateralises the inverse short? This is a standard cash-and-carry mechanism; the research contribution is its tested accounting, execution and no-trade rule.

The mathematical identity

Let C be dollar face, F0 the futures entry price, Ft the futures mark and St spot. A short inverse contract earns coin P&L C(1/Ft - 1/F0). Buy net coin margin q = C/F0 after paying the entry coin fee.

Step	Identity before later fees
Coin equity	q + C(1/Ft - 1/F0) = C/Ft
Value in spot currency	Vt = C St/Ft
At ideal expiry convergence	If FT = ST, value becomes C.

In the real test, exits occur before delivery at actual recorded prices. Residual basis is retained. Each entry slice is matched separately; partial closes release only matching collateral. Integer contract faces are USD 100 for BTC and USD 10 for ETH. Spot lots round to 0.001 coin; unsold dust remains marked inventory.

Fixed execution and selection rules

Test all 44 BTC/ETH quarterly contracts from 2021Q1 through 2026Q2. Decide 60 days before expiry from a completed prior-hour close. Use 24 hourly entry slices and 24 exit slices ending one hour before delivery. Every slice is capped at 1% of prior 24-hour mean hourly notional volume on both legs.

Allocate a 40% cash budget per asset. The gate requires the prior observed premium to exceed four-leg base costs, cash opportunity cost and a 2% annual risk buffer. Rejected slots stay cash. No horizon, asset or threshold was optimised on later returns.

A real capacity limit

The first USD 1 million development simulation failed its scheduled unwind by 9 ETH contracts (USD 90 face). That failure is retained. The supported implementation is a USD 100,000 proof-of-concept, under the same dates, thresholds and participation limits. It does not establish million-dollar capacity.

4 | Funding: forecast cash flow, reserve capital

The matched linear hedge

Buy q units of spot and short exactly q units of a linear perpetual. Price P&L is q[(ST - S0) - (FT - F0)]. Add observed funding q x settlement mark x funding rate and subtract both-leg costs. Funding is paid only on inventory held before the event; the current payment cannot be used to enter and earn itself.

Three declared models

Model	Rule
Always on	Continuously hold the matched spot/short hedge.
EWMA gate	Conservative minimum of past 7-day and 30-day funding estimates.
Ridge gate	Forecast next 30-day funding using lagged funding, basis, returns and volatility; alpha = 10.

Ridge uses quarterly trailing 730-day training, training-only standardisation and a 30-day label purge. Every training target must end before the fit date. Daily decisions use completed information and execute one hour later. A three-day weak-funding rule, adverse-basis exit and weekly integer-lot resizing are explicit.

Collateral is not free

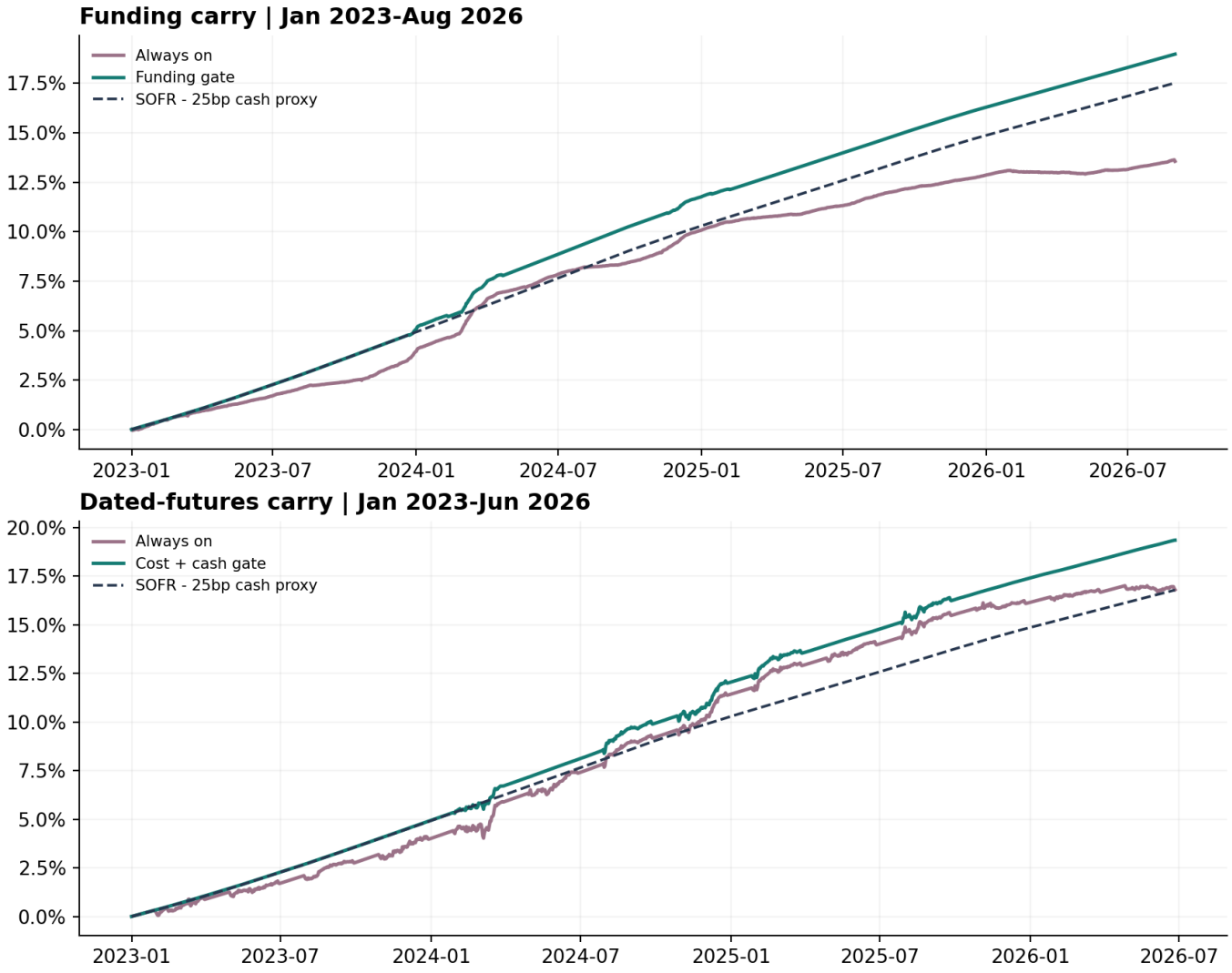
Target 20% NAV spot per asset plus fully reserved short margin. Keep futures unrealised P&L separate from physical cash. Reserve extra cash for unrealised short losses; posted reserves earn no interest. If falling spot leaves cash short despite offsetting futures gains, realise those gains through a paid futures close/reopen. No synthetic free cash is created.

Selection and research history

Original 2022 validation selected EWMA. After later preliminary results were seen, an exit-dust execution bug and the coarse cash proxy were corrected uniformly. The original selection was retained. These are audit-corrected retrospective chronological results, not a pristine unseen holdout.

The final later-period funding result has 6 holding episodes. Both single-asset removal checks retain positive base-cost excess return, with unused slots held in cash. No winning asset is reselected from those diagnostics.

5 | Returns after execution and cash costs

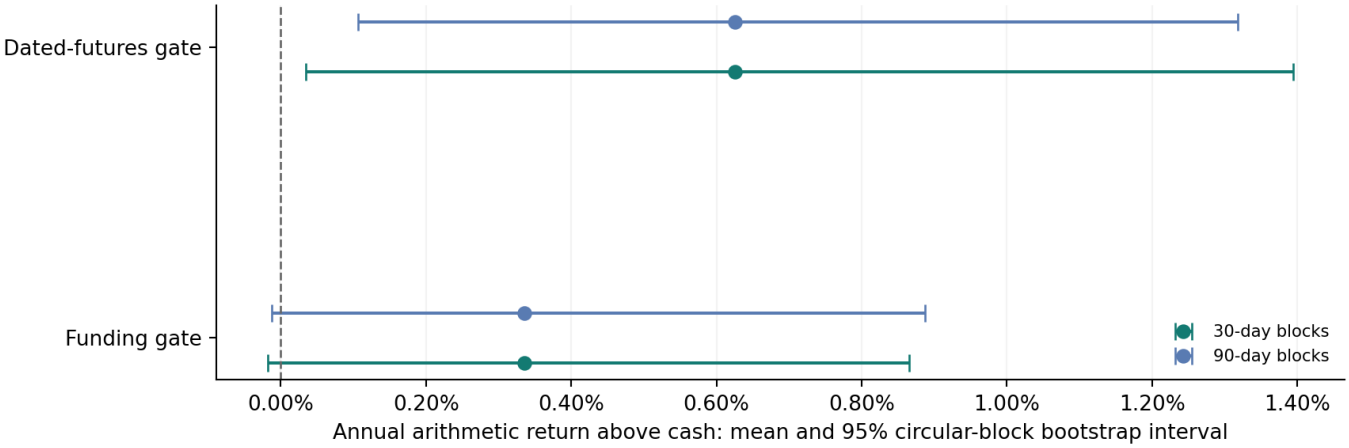


Cost sensitivity: spot / futures, one-way bp

Design	Cost bp	Net total	Cash total
Funding	12/7	18.97%	17.52%
Funding	20/12	18.53%	17.52%
Funding	30/20	17.91%	17.52%
Dated	12/7	19.35%	16.79%
Dated	20/12	18.28%	16.79%
Dated	30/20	16.79%	16.79%

Costs combine fees and adverse execution. They are not measured historical bid/ask spreads. The base-cost entry hurdle stays unchanged in stress runs. Severe costs leave very little dated-futures excess return; a favourable base case is not a universal profitability claim.

6 | Uncertainty and economic significance



Design / block	Annual excess mean	95% interval	LRV Sharpe
Funding / 30d	0.34%	[-0.02%, 0.86%]	0.76
Funding / 90d	0.34%	[-0.01%, 0.89%]	0.72
Dated / 30d	0.63%	[0.03%, 1.39%]	0.99
Dated / 90d	0.63%	[0.11%, 1.32%]	1.07

LRV uses Bartlett HAC long-run variance at 30/90 daily lags. Intervals use 2,000 circular-block bootstrap samples. These diagnostics preserve some serial dependence; they do not create independent trades, remove all research-selection bias or price exchange-default risk.

Where the money came from

Funding gate: 189,725 USDT total P&L = 36,686 funding + 157,426 cash interest + 1,544 basis/execution P&L - 5,930 costs. Most of the headline return is cash interest. Excess final wealth versus the cash proxy is 14,544 USDT.

The dated gate accepted 9 of 28 later asset-expiry decisions. The BTC and ETH positions in the same quarter are related observations. Later daily max drawdown is -0.37%, while the full 2021-onward path includes the larger 2021 basis-widening drawdown (about 2.65%).

Unpriced loss can overwhelm the edge

A hypothetical 2% loss on the funding strategy's peak exchange receivable is about 19,166 USDT, compared with 14,544 excess final wealth. This is a loss scenario, not an estimated default probability. A smooth price ledger is not a complete risk model.

7 | Data integrity and reproducibility

What has been verified

31 meaningful tests passed: hedge identities, funding ownership/timing, cash and coin fee accounting, partial unwinds, integer-lot exits, label purging, future-data invariance, unavailable bars and publication-lag/day-count handling. All 15 final ledgers reconcile NAV, order fees and information-before-execution. No declared hourly margin-stress breach occurred.

962 source archives match their retained download hashes: 640 monthly spot/perpetual/funding archives, 264 dated-futures monthly archives and 58 daily repairs. The package includes raw data, processed inputs, source URLs, protocols, fit logs, final ledgers and tests. Downloaded copyrighted research-paper PDFs are excluded.

Data limitations that remain

There are 31 unavailable spot-print hours per asset over 2020-Aug 2026. A flagged previous spot/mark basis proxy is used only to value inventory and cannot fill an order. Dated-contract gaps were recovered from official daily archives. A missing composite-index source was not substituted as observed data.

The USD futures / USDT spot comparison assumes parity. Cash interest is an assumed off-exchange SOFR-minus-25bp proxy with a seven-day information lag, not an investable fund total-return series. Transfers, historical depth, queues, asynchronous legs, ADL, taxes and exchange failures are not reconstructed. Hourly OHLC stress is conditional evidence, not a general liquidation guarantee.

Reproduce from the included snapshot

• Install the pinned requirements in a fresh Python environment.

• Run python -m pytest -q tests.

• Run carry.py --phase development and carry.py --phase locked. The latter checks the frozen code/protocol hashes.

• Run inverse.py, robustness.py and diagnostics.py. Read the final summaries rather than superseded audit folders.

The next experiment

Freeze these rules and record forward bid/ask, available depth, actual fees, quote-to-fill slippage, rejected orders and transfer timing. Keep an independent paper ledger and test a second venue without choosing it for historical returns. Better execution evidence and more independent opportunities take priority over more ML or leverage.

8 | Turning the work into a quant CV project

Suggested title

Cost-Aware Relative-Value Trading Research

Evidence-backed bullets

• Derived matched spot/inverse-futures hedges and tested 44 BTC/ETH quarterly contracts with coin fees, integer sizing and volume-limited execution.

• A fixed cost-and-cash filter raised 2023-Jun 2026 net return to 19.35% from 16.81% always-on, reducing trading costs by 67%.

• Built an audited hourly research engine with causal signals, collateral accounting, cost stresses and 31 tests; assessed results using block bootstrap and HAC inference.

Recruiter and technical-review assessment

Dimension	Assessment
Research question	Clear economic mechanism, rather than a generic clustering exercise.
Mathematics	Exact inverse-contract hedge, state-space comparison and explicit capital accounting.
Engineering	Reproducible public-data pipeline, source integrity, event-time ledgers and meaningful tests.
Empirical result	Positive under stated assumptions; modest excess and limited independent episodes.
Novelty	A careful implementation and comparison of known mechanisms, not a new arbitrage theorem.

This is stronger than the old project because its numbers can be traced to cash flows and its failure cases are documented. Do not advertise a guaranteed strategy, a novel pricing result, million-dollar capacity, or the unadjusted funding Sharpe as the main achievement.

Before an interview, be able to derive the coin hedge, explain why the ETF version failed, reproduce one trade by hand, reconcile fees and collateral, and explain why cash, capacity and dependence change the result.

References and primary data sources

1. Gatev, Goetzmann & Rouwenhorst (2006) Original source. Access: Abstract/working-paper summary.
2. Avellaneda & Lee (2010; 2009 manuscript) Original source. Access: Full author manuscript.
3. Clegg & Krauss (2018) Original source. Access: Abstract + author implementation.
4. Leung & Li (2015) Original source. Access: Full author preprint.
5. Rad, Low & Faff (2016) Original source. Access: Full accepted manuscript.
6. Bailey & Lopez de Prado (2014) Original source. Access: Full author manuscript.
7. Engelberg, Gao & Jagannathan Original source. Access: Abstract only.
8. Schmeling, Schrimpf & Todorov (BIS, 2023) Original source. Access: Full BIS paper.
9. He, Manela, Ross & von Wachter Original source. Access: Full-text preprint HTML.
10. Ackerer, Hugonnier & Jermann Original source. Access: Full-text preprint HTML.
11. Lo (2002) Original source. Access: Publication abstract.

Contract and data documentation

Official Binance public archive documentation

Coin-margined contract faces and collateral

Inverse-contract P&L formulas

Quarterly delivery and settlement convention

NYFed SOFR definition and publication

Kenneth French factor data (ETF pilot)

Interpretation boundary

Literature motivates the hypotheses; it does not validate this implementation. All performance numbers in this report come from the accompanying code and input snapshot, not from published paper returns. Source-access levels are recorded in the separate review. The study includes unsuccessful branches and post-observation corrections so that later readers can assess research selection honestly.